

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

6. Let $C = A \times B$, with $A = [1, 0, 1, 0]$ and $B = [0, 0, 0, 1]$ given as lists of bits, in increasing order of significance, that is: lowest bit first, highest bit last.
- (a) Apply Karatsuba's algorithm to compute C . Show all steps.
 - (b) Give the recurrence relation which justifies the running time.
Explain the recurrence by referring to the example.

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Answer:

6. (a) Application of Karatsuba's algorithm.
RM abbreviates the Recursive-Multiply function.
- i. $S_0 = [1, 0] + [1, 0] = [0, 1]$, $S_1 = [0, 1] + [0, 0] = [0, 1]$
Three recursive calls: $c_2 = \text{RM}([1, 0], [0, 1], 2)$, $c_0 = \text{RM}([1, 0], [0, 0], 2)$, and
 $c_1 = \text{RM}([0, 1], [0, 1], 2) - c_0 - c_2$.
 - ii. The first two recursive calls reduce to the base case in a straightforward manner and give $c_2 = [0, 1]$ and $c_0 = [0, 0]$.
The third recursive call requires one shift and an extra addition, which leads to
 $c_1 = [0, 0, 1] - c_0 - c_2 = [0, 1]$.
 - iii. Assembling leads to $[0, 0, 0, 1, 0, 1]$.
- (b) The recurrence is $T(n) = 3T(n/2) + O(n)$ which resolves to $O(n^{1.59})$.
In the example we execute 3 recursive calls on problems that are half the size, which corresponds to the $3T(n/2)$ term in the recurrence.
The $O(n)$ term in the recurrence is explained by the linear cost of the addition, the shift, and the assembling of the result.